

Forensics for Cybercrimes Committed Using Artificial Intelligence

A Digital Forensic Framework for AI-Assisted Cybercrime

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2026

Research Context

Artificial Intelligence is changing the cybercrime landscape.

Traditional Cybercrime

- ▶ Human-controlled systems
- ▶ Identifiable devices
- ▶ Conventional malware
- ▶ Established forensic procedures

AI-Assisted Cybercrime

- ▶ Automated attacks
- ▶ Synthetic identities
- ▶ AI-generated content
- ▶ Autonomous decision-making

The Forensic Challenge



What happened?



Who or what caused it?



Can we prove it?

Formalising Digital Evidence

Let

$$E = \{e_1, e_2, \dots, e_n\}$$

represent the set of collected digital evidence.

Let

$$A \subseteq E$$

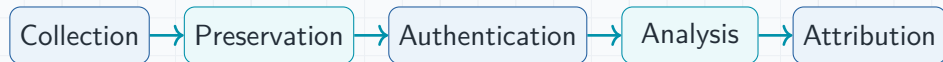
represent evidence that satisfies the authentication criteria.

Therefore:

$$\boxed{A \subseteq E}$$

and

Forensic Evidence Pipeline



Core Principle

Evidence must remain traceable from collection through analysis and ultimately to attribution and presentation.

Research Objectives

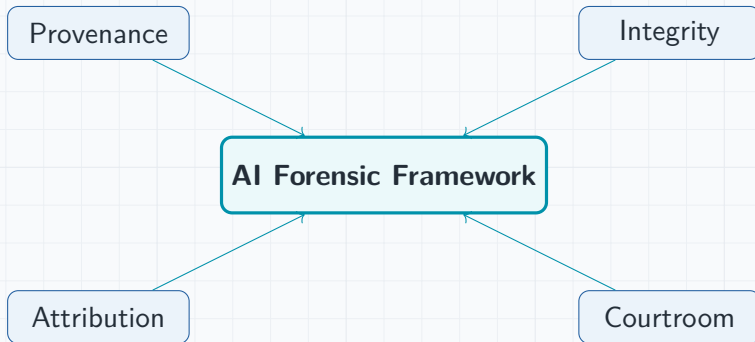
General Objective

To develop a digital forensic framework for investigating cybercrimes committed using Artificial Intelligence.

Specific Objectives

1. Examine AI-assisted cybercrime techniques.
2. Evaluate existing digital forensic approaches.
3. Investigate evidence provenance and integrity.
4. Develop a forensic framework.
5. Evaluate the proposed framework.

Expected Contribution



Conclusion

Central Argument

AI changes not only how cybercrimes can be committed, but also how digital evidence must be interpreted, authenticated and attributed.

AI + Cybercrime + Digital Forensics + Evidence

Thank You

Questions & Discussion

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